

Type Deligne - Lusztig varieties and type sheaves

Lecture 1 (Charlotte Chan)

1976, Deligne - Lusztig,

G conn. red. gp / $\mathbb{F}_q$, $\Pi \hookrightarrow G$ max'l form, $\sigma = \text{assoc. Frob}$

Def. For $B \subset G_{\overline{\mathbb{F}}_q}$ containing $\Pi_{\overline{\mathbb{F}}_q}$,

$$\begin{array}{ccc} & & R_u(B) \\ & & \uparrow \\ X_{0,0+}^{(\Pi \subset G, B)} & := & \{ g \in G_{\overline{\mathbb{F}}_q} : g^{-1} \sigma(g) \in U \} \\ \uparrow & & \uparrow \\ G(\mathbb{F}_q) & & \Pi(\mathbb{F}_q) \end{array}$$

$$\theta: \Pi(\mathbb{F}_q) \rightarrow \mathbb{C}^\times, \quad R_{0,0+}^{(\Pi \subset G, B)}(\theta) := \sum_{i=0}^{\infty} (-1)^i H_c^i(X_{0,0+}^{(\Pi \subset G, B)})_\theta$$

Ex. $G = SL_2$, $\Pi = \begin{pmatrix} * & * \\ & * \end{pmatrix}$, $\sigma \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^q & b^q \\ c^q & d^q \end{pmatrix}$, $B = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$

$$\sim X_{0,0+}/U = G(\mathbb{F}_q)/U(\mathbb{F}_q)$$

$$\rightarrow R_{0,0+}(\theta) = \text{Ind}_{B(\mathbb{F}_q)}^{G(\mathbb{F}_q)} (\theta \circ \text{pr})$$

$$\theta: \left(\begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} \right) \rightarrow \mathbb{C}^\times$$

$a \in \mathbb{F}_q^\times$

- $G = SL_2$, $\Pi = \begin{pmatrix} * & * \\ & * \end{pmatrix}$, $\sigma \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d^q & c^q \\ b^q & a^q \end{pmatrix}$

$$\rightarrow X_{0,0+} \simeq V(x^{q+1} - y^{q+1} = 1)$$

$$SL_2(\mathbb{F}_q) = \left\{ \begin{pmatrix} a & b^q \\ b & a^q \end{pmatrix} : a, b \in \mathbb{F}_{q^2}, \det = 1 \right\}$$

$$\Pi(\mathbb{F}_q) = \left\{ \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} : a \in \mathbb{F}_{q^2}^\times, \det = 1 \right\} \simeq \mu_{q+1}$$

Th. (Deligne-Lusztig)

$$\left\langle R_{0,0+}^{(\pi \subset G, B)}(\theta), R_{0,0+}^{(\pi' \subset G, B')}(\theta') \right\rangle_{G(\mathbb{F}_q)} = \sum_{w \in W_{G(\mathbb{F}_q)}(\pi, \pi')} \langle \theta^w, \theta' \rangle_{\pi'(\mathbb{F}_q)}$$

In particular,

- $R_{0,0+}^{(\pi \subset G, B)}(\theta)$ is indep. of choice of B .

- $R_{0,0+}^{(\pi \subset G, B)}(\theta)$ is irred. $\Leftrightarrow \text{Stab}_w(\theta) = \{1\}$.

2004, Lusztig Q: $SL_2(\mathbb{F}_q(\epsilon)/\epsilon^{r+1})$

Let G conn. red. gp / F has local

$T \hookrightarrow G$ max'l torus, split over F^{ur}

$$x \in A(T) \cap B(G, F)$$

Def. For $B \subset G^{\text{ur}}$

$$X_{0,r+}^{(\pi \subset G, B)} = \left\{ g \in (G_{0,r+})_{\overline{\mathbb{F}_q}} : g^{-1} \sigma(g) \in U_{0,r+} \right\}$$

$$\text{where } G_{0,r+} := L^+ g_{x,0} / L^+ g_{x,r+}$$

& def. $R_{0,r+}^{(\pi \subset G, B)}(\theta)$ analogously.

$$\text{Ex. } G = SL_2, \quad G_{0,r+}(A) = \begin{cases} SL_2(A[\epsilon]/\epsilon^{r+1}) \\ A \text{ } \mathbb{F}_q\text{-alg} \\ \text{perfect} \end{cases} \quad U_{0,r+}(A)$$

$$a) X_{0,r+}/U_{0,r+} = G_{0,r+}(\mathbb{F}_q)/U_{0,r+}(\mathbb{F}_q)$$

$$R_{0,r+}(\theta) = \text{Ind}_{B_{0,r+}(\mathbb{F}_q)}^{G_{0,r+}(\mathbb{F}_q)} (\theta \circ \text{pr})$$

$$b) X_{0,r+1} = \mathbb{V} \left(\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{F}_q[\epsilon]/\epsilon^{r+1} : \det \begin{pmatrix} x & \sigma(y) \\ y & \sigma(x) \end{pmatrix} = 1 \right)$$

Th. (2004 Lusztig, 2009 Starasinski, 2021 C. - Ivanov)

If θ is generic in the sense that $\theta|_{\pi_{r,r+1}(\mathbb{F}_q) =: \mathbb{L}}$ restricts nontrivially to each $\mathbb{L}_\alpha$ then the scalar prod. formula holds.

Q. What if θ is not generic?

Def. $(T \subset G, \theta, x) \xrightarrow[\text{(Taletha)}]{\text{Hove factorization}} (\vec{G}, \vec{\Phi}, \vec{r}, x) =: \mathbb{F}$

For $B \subset G_{\text{Fur}}$ cont. T_{Fur}

$$R_{0,r+1}^{(\mathbb{F}, B)}(\theta) = \dots \left(\text{Inf} \left(R_{0,r+1}^{(G^0, C^0)} \left(\text{Inf}_{G_{0,r+1}}^{G_{0,r+1}} \left(\text{Inf}_{G_{0,r+1}}^{G_{0,r+1}} R_{0,r+1}(\phi_{-1}) \otimes \phi_0 \right) \right) \otimes \phi_1 \right) \right) \dots$$

Rem. CS version of this works too.

$(T, I_\theta, x) \rightsquigarrow \exists \theta$ simple perverse sheaf on $G_{0,r+1}$

Th. (C., 2024) For any B, θ , the scalar prod. formula holds for $R_{0,r+1}^{(\mathbb{F}, B)}(\theta)$.

In particular, - indep of B

- $R_{0,r+1}^{(\mathbb{F}, B)}(\theta)$ is irred. $\Leftrightarrow \text{Stab}_{W_{G_{0,r+1}}(\mathbb{F}_q)}(T_{0,r+1})(\theta) = \{1\}$.

Q. What are these reps? eg. compare to χ_u ?

"Isomorphisms on sale" π_1, π_2 irrep, $\pi_3 \simeq \pi_2 \Leftrightarrow \oplus \pi_3|_A = \oplus \pi_2|_A$.

Ex. $G = \mathbb{Z}/N\mathbb{Z}$, $\pi_1 \simeq \pi_2 \Leftrightarrow \textcircled{H} \pi_1(1) = \textcircled{H} \pi_2(1)$

Ex. $H = \left\{ \begin{pmatrix} 1 & a & c \\ & 1 & b \\ & & 1 \end{pmatrix} : a, b, c \in \mathbb{F}_p \right\}$ Heisenberg gp

π_1, π_2 not 1-dim'l, $\pi_1 \simeq \pi_2 \Leftrightarrow \textcircled{H} \pi_1 \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \textcircled{H} \pi_2 \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(Lusztig's 1977, (-oi 2023)) θ in gen. position

Th. Let π irrep of $G(\mathbb{F}_q)$. Assume $q \gg 0$, then

Henriart: $h_n(F)$, s.c.

(-oi: $h(F)$)

(-oi: $h(F)_{x,0}$)

$\pi \simeq R_{0,0+}(\theta) \Leftrightarrow \textcircled{H} \pi|_{G(\mathbb{F}_q)_{\text{rss}}} = \textcircled{H} R_{0,0+}(\theta)|_{G(\mathbb{F}_q)_{\text{rss}}}$

||
 π'

pb

Goal. $\langle \pi, \pi' \rangle = 1$. ETS: $\langle \pi, \pi' \rangle \neq 0$

Cauchy-Schwarz: $|\langle \pi, \pi' \rangle_{\text{not}}| \leq \langle \pi, \pi \rangle_{\text{not}}^{1/2} \cdot \langle \pi', \pi' \rangle_{\text{not}}^{1/2} = \langle \pi, \pi \rangle_{\text{not}}$

$\langle \pi, \pi \rangle = \langle \pi, \pi \rangle_{\text{rss}} + \langle \pi, \pi \rangle_{\text{not}}$

||
 $\langle \pi', \pi' \rangle = \langle \pi', \pi' \rangle_{\text{rss}} + \langle \pi', \pi' \rangle_{\text{not}}$

$\therefore$ ETS $\langle \pi, \pi \rangle_{\text{not}} < \frac{1}{2}$. (why? $\langle \pi, \pi' \rangle_{\text{rss}} + \langle \pi, \pi' \rangle_{\text{not}} \neq 0$)

$\therefore$ ETS $\langle \pi, \pi \rangle_{\text{rss}} > \frac{1}{2}$ | | $> \frac{1}{2}$ $< \frac{1}{2}$

(calculate: if $\frac{|\mathbb{F}(\mathbb{F}_q)|}{|\mathbb{F}(\mathbb{F}_q)_{\text{not}}|} > 2|w|$)

Th (-oi, 2025) Assume $q \gg 0$ so that isom. on sale holds.

Then $R_{0,0+}^{(\mathbb{F}, \mathbb{B})}(\theta) = \text{Kim-Yu type assoc. to } (T \in G, \theta, \mathbb{F})$.

pb - Use isom on sale if $\text{Stab}_w(\theta) = \{1\}$

- Green function trick.

Th (Lipman) True for $q \gg 0 \Rightarrow$ true $\forall q$

Pt Shintani descent.

Big Q T not unram?

Def (CKZ) Assoc to $(T \hookrightarrow G, \theta, x)$, (can define
family ram. $\nearrow$ $A(T) \cap B(G, F)$)

- type DL var. $X_{0, r+}^{(\mathbb{F}, B)}$

- type DL ind: $R_{0, r+}^{(\mathbb{F}, B)}(\theta)$.

Th (C. - Kaelin-Letzer - Zhu) $R_{0, r+}^{(\mathbb{F}, B)}(\theta) =$ Kim-Yu type assoc. to $(T \hookrightarrow G, \theta, \xi)$

Rem. - Iwanari - Nie use the defn to obtain this result for $q \gg 0$.
 $\nwarrow$
T elliptic case

- Yuta Takeda independently has nice work for a special case of $X_{0, r+}^{(\mathbb{F}, B)}$

Lecture 2 (TK) Towards type Deligne-Lusztig theory.

F na local field,
tame twisted Levi's

G/F conn reductive $p \nmid |W|$
Kim, Fintzen

depth 0
irr. s.c.
rep of $G^0(F)$ $\left\{ \begin{array}{l} G^0 \subset \dots \subset G^d \\ \pi_{i-1} \\ \phi_0, \dots, \phi_d \end{array} \right\}$

$\xrightarrow{\text{Yu}}$ $\left\{ \begin{array}{l} \text{irred s.c.} \\ \text{rep of } G(F) \end{array} \right\}$

$\phi_i: G^i(F) \rightarrow \mathbb{C}^\times$
 G^{i+1} - generic

fibres Hakim-Monaghan
"refactorization"

Def Let $S \subset G$ tame max'l torus

$\theta: S(F) \rightarrow \mathbb{C}^\times$. A Hone factorization is

$S = G^{-1} \subset G^0 \subset \dots \subset G^d = G$ tame twisted Levi

$\phi_{-1} \quad \phi_0 \quad \phi_d \quad \phi_i: G^i(F) \rightarrow \mathbb{C}^\times$

$$1) \prod_{i=-1}^d \phi_i|_{S(F)} = \theta$$

2) ϕ_i is G^{i+1} -generic, $i=0, \dots, d-1$

3) $\phi_i|_{G_{S_0}^i(F)} = 1$ $\leftarrow$ normalized

Th. 1) Any pair (S, θ) has a Hone factorisation

2) Any two such differ by a refactorisation

Th. Hone factorisation gives a bijection

1) $G(F)$ -conj. classes of (S, θ) , $S \subset G$ tame ell. max. torus

$\theta: S(F) \rightarrow \mathbb{C}^\times$ regular, $S \subset G^0$ max unramified

2) reg s.c. irreps

$$\pi \text{ reg} \Leftrightarrow \pi|_S \text{ reg} \Leftrightarrow \pi|_S = (-\text{ind } R_T(\phi_{-1}))$$

$\uparrow$
 $\phi_{-1} \text{ reg.}$

cf. Th (Harish-Chandra) $G/\mathbb{R}$. $G(\mathbb{R})$ -conj. classes (S, θ) , $S \subset G$ ell max

$\uparrow$
irred (reg) d.s. reps of $G(\mathbb{R})$

$\theta: S(\mathbb{R})_\pm \rightarrow \mathbb{C}^\times$
reg diff.

Characters

π admissible

$$\Theta_{\pi}^{\text{dist}}(f) = \text{tr} \left(v \mapsto \int_G f(g) \pi(g) v dg \right)$$

$$\Theta_{\pi} : \mathcal{L}(F)_{\text{rs}} \rightarrow \mathbb{C}, \quad \Theta_{\pi}^{\text{dist}}(f) = \int_G f(g) \Theta_{\pi}(g) dg$$

$$D_G(g) = \left| \det(1 - g|_{\text{Lie } \mathfrak{a} / \text{Lie } \mathfrak{a}_g}) \right|$$

Th. Let $\gamma \in S(F)$ reg. top. s.s.

$$D_G^{-1} \Theta_{\pi}(\gamma) = e(u) \varepsilon\left(\frac{1}{2}, X^*(T_0)_{\mathbb{C}} - X^*(S)_{\mathbb{C}}, \wedge\right)$$

$$\sum_w \Delta_{\Pi}(\gamma^w) \boxed{\varepsilon(\gamma^w)} \Theta(\gamma^w)$$

Th (H-C) $\mathbb{C}/\mathbb{R}$, $\gamma \in S(\mathbb{R})$ reg

$$D_G^{-1} \Theta_{\pi}(\gamma) = e(u) \varepsilon\left(\frac{1}{2}, X^*(T_0)_{\mathbb{C}} - X^*(S)_{\mathbb{C}}, \wedge\right)$$

$$\sum_w \Delta_{\Pi}(\gamma^w) \Theta(\gamma^w).$$

Twist γ_u construction by a character $\varepsilon : G^*(F)_x \rightarrow \{\pm 1\}$

Q. Can we find a geometric interpretation of γ_u 's construction, and of ε .

Recall classical DL theory

k finite, G/k conn. red. $S \subset G$ max k -torus, $S_{\bar{k}} \subset B \subset G_{\bar{k}}$
 $\begin{matrix} S_{\bar{k}} \cdot U \\ \parallel \\ B_{\text{red}} \end{matrix}$

$$Y_B^G = \{g \in G/U : g^{-1} \sigma(g) \in U \sigma(u)\}$$

$$\begin{matrix} \nearrow \\ G(k) \end{matrix} \hookrightarrow S(k)$$

$$G(k) \simeq \text{It}_G^{\vee}(Y_B^G, \bar{a}_e)_{\Theta}, \quad \Theta : S(k) \rightarrow \bar{a}_e^{\vee}$$

$$R_S^G(\theta) := \sum_{i=0}^{\infty} (-1)^i H_c^i(Y_B^G, \bar{Q}_\theta)_\theta$$

If θ is regular,

$$1) H_c^i(Y_B^G, \bar{Q}_\theta)_\theta = 0 \quad \text{if } i \neq \dim(Y_B^G)$$

$$2) H_c^i(Y_B^G, \bar{Q}_\theta)_\theta \text{ is irreducible} \quad i = \dim(Y_B^G)$$

$$\dim(Y_B^G) = \dim(U/U \cap \sigma(U))$$

$$B \longleftrightarrow R^+ \subset R(S, G) \subset X^*(S) \cap \Sigma = \Gamma \times \{\pm 1\}$$

Heisenberg - Deligne - Lusztig variety (HDL)

$$k = \mathbb{F}_q, \quad p \neq 2$$

(V, η) symplectic vector space / k

$$H \cong a_\alpha \boxtimes_\eta V : (a, v)(b, w) = (a + b + \eta(v, w), v + w)$$

Th (Stone-von-Neumann) For every $1 \neq \psi: k \rightarrow \bar{\mathbb{A}}_k^\times$, there is a unique irrep Heis_ψ of $H(k)$ whose central char is ψ .

In fact, $\text{Heis}_\psi = \text{Ind}_{a_\alpha(k) \times L(k)}^{H(k)} (\psi \boxtimes \mathbb{1})$, $L \subset V$ Lagrangian

Th (Weil) There is a unique ext of Heis_ψ to a rep WeilHeis_ψ of $H(k) \rtimes \text{Sp}(V, \eta)(k)$.

Def (HDL) Choose $L \subset V_{\bar{k}}$. Define

$$\sim \quad Y_L^H := \left\{ h \in H/L : h^{-1} \sigma(h) \in L \cdot \sigma(L) \right\}$$

$\hookrightarrow G_a(k)$

Prop 1) $\dim(Y_L^H) = \dim(L / L \cap \sigma(L))$

2) Assume $1 \neq \psi: G_a(k) \rightarrow \bar{G}_e^X$. Then

$$H_c^i(Y_L^H, \bar{G}_e)_{\psi} = \begin{cases} 0, & i \neq \dim(Y_L^H) \\ \text{Heis}_{\psi}, & i = \dim(Y_L^H) \end{cases}$$

3) Let $S \subset Sp(V, \eta)$ be a max. k -torus. Normalizing L , Then

$$H(k) \rtimes S(k) \sim H_c^i(Y_L^H, \bar{G}_e)_{\psi} = \text{WeilHeis}_{\psi} \Big|_{H(k) \rtimes S(k)} \otimes (1 \boxtimes \xi_S)$$

where $\xi_S: S(k) \rightarrow \{\pm 1\}$ related to ξ .

Analogies. 1) Formula

1) Borel B containing $S_{\bar{k}} \Leftrightarrow B$ normalized by S

$$\Leftrightarrow R^+ \subset R(S, \eta)$$

(set of reps for $\{\pm 1\}$ -orbits)

Lagrangian normalized by $S \Leftrightarrow \Xi^+ \subset \Xi = \text{weight of } S \text{ on } V$

1) dim formula same.

reps for $\{\pm 1\}$ -orbits

Proof in case S split, classical DL: $Y_B^G = G(k)/U(k)$

$$H_c^0(G(k)/U(k), \bar{G}_e)_{\theta} \simeq \text{Ind}_{S(k) \rtimes U(k)}^{G(k)} (\theta \boxtimes \mathbb{1})$$

HDL $\gamma_L^H = H(k)/L(k)$

$$H_c^0(H(k)/L(k), \bar{\psi})_\psi = \text{Ind}_{G_a(k) \times L(k)}^{H(k)} (\psi \otimes 1)$$

$$= \text{Ind}_{(G_a(k) \times L(k)) \times S(k)}^{H(k) \times S(k)} (\psi \otimes 1 \otimes 1)$$

$$= \text{WeilHeis}_\psi \Big|_{H(k) \times S(k)} \otimes \left(1 \otimes \text{sgn det}(-|L(k)) \right)$$

$P \subset Sp(V, \eta)$ the Siegel parabolic for L , $\text{Stab}_{Sp}(L)$

$$\text{Ind}_{(G_a(k) \times L(k)) \times P(k)}^{H(k) \times P(k)} (\psi \otimes 1 \otimes 1) = \text{WeilHeis}_\psi \Big|_{H(k) \times P(k)} \otimes \left(1 \otimes \text{sgn det}(-|L(k)) \right)$$

Proof when S is anisotropic, split over k_2/k

Then L rat'l over k_2 , $V_{k_2} = L_{k_2} \oplus \sigma(L_{k_2})$

Define $\bar{\gamma}_L^H = \gamma_L^H / G_a(k)$, $\bar{\gamma}_L^H = \{v \in V/L : \sigma(v) - v \in L + \sigma(L)\} = V/L$

$\gamma_L^H \rightarrow \bar{\gamma}_L^H$ $G_a(k)$ -torsion
 $\downarrow$
 L_ψ local system

$$H_c^1(\gamma_L^H, \bar{\psi})_\psi = H_c^1(\bar{\gamma}_L^H, L_\psi)$$

We can find a decomp
 $\dim V/2$
 $V = \bigoplus_{j=1}^{\dim V/2} V_j$, $\dim(V_j(k) \cap L(k))$
 $= \dim(V_j(k) \cap \sigma(L)(k)) = 1$

$$\bar{\gamma}_L^H = \prod \bar{\gamma}_{L_j}^H, \quad L_\psi = L_{\psi_1} \otimes \dots \otimes L_{\psi_r}$$

K nneth $\rightarrow \dim V = 2, \quad V = \bar{k} \oplus \bar{k}, \quad \eta = x_1 y_2 - x_2 y_1,$

$$S = \begin{bmatrix} * & \\ & * \end{bmatrix}$$

$$Sp(V) = SL_2$$

$$L = \bar{k} \oplus \{0\}$$

$$\sigma = \text{Ad} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \circ \sigma_{\text{std}}$$

$$Y_L^H = \left\{ (a, x) \in \bar{k} \times \bar{k} : a^q - a = -x q^2 + q \right\}$$

$$\begin{array}{c} \downarrow \quad \downarrow \\ \overline{Y_L^H} = \bar{k} \end{array}$$

$$H_c^0(V/L, L_4) = 0$$

$$H_c^2(V/L, L_4) = 0$$

On $H_c^1(V/L, L_4)$, $\text{Gal}(\bar{k})$ acts by ψ .

$$\chi_c(L_4) = 1 - (q+1) = -q$$

$$\text{sw}(\omega) = q+1$$

Lecture 3 (XZ) Weil-Heisenberg-Deligne-Lur  ig var

Recall. $(T \hookrightarrow G, x, \phi: T(F) \rightarrow \overline{G}_e^x)$

$$\phi = \phi_{-1} \pi \phi_1$$

$$T \hookrightarrow G^0 \hookrightarrow G^1 \hookrightarrow \dots \hookrightarrow G^d, \quad 0 < r_1 < r_2 < \dots < r_{d-1} \leq r_d$$

$$s_i = \frac{r_i}{2}$$

$$T_{x,0} \hookrightarrow G_{x,0}^i$$

$$K = \prod_{i=0}^{d-1} G_{x,s_i}^i$$

$$T_{x,0+} \twoheadrightarrow \mathbb{Z}$$

$$\begin{array}{c} \nabla \\ K_1 = G_{x,0+}^0 \prod_{i=1}^{d-1} G_{x,s_i}^i \end{array} \left. \begin{array}{c} \text{quotient } \overline{G_{x,0}^0} \\ \text{quotient } V \end{array} \right) \begin{array}{c} \mathbb{Z} \\ \hat{\phi}: K_2 \rightarrow \overline{G}_e^x \\ \text{quotient } \overline{G_{x,s_i+}^i} \\ \text{quotient } V \end{array} \begin{array}{c} \hat{\phi}: K_2 \rightarrow \overline{G}_e^x \\ K_3 = \ker(\hat{\phi}) \\ \uparrow T_{x,0+} \end{array}$$

equal char
 $M \propto \underbrace{(V \times Z)}_H$

Idea: construct WH reps using
 not nec. rat'l Lagrangians

M may not fix a Lagrangian in V

Σ

General setup $K = \mathbb{F}_q, \bar{K} = \overline{\mathbb{F}_q}$

(V, η) symplectic vector space / K

$H = G_a \times V, (a, v)(a', v') = (a + a' + \eta(v, v'), v + v')$, L Lagrangian $\subset V$
 $\hat{L} = G_a \times L \subset H$ subgp

M (cont'd) reductive group $\rightarrow Sp(V, \eta)$

$\curvearrowright$

$H = H \rtimes M$

Fix $1 \neq \psi: G_a(K) = K \rightarrow \bar{G}_a^*$

WH Ψ Weil-Heisenberg rep of $h(K)$

Fix S_M max'l torus of M/K .

Fix a Lagrangian $L/\bar{K} \subset V_{\bar{K}}$. $B_M/\bar{K} \subset M_{\bar{K}}$

$S_M \subset B_M, B_M$ fixes L

$B = \hat{L} \rtimes B_M \hookrightarrow S = G_a \times S_M$

$\cup$

$U = L \rtimes U_M$

$Y_{SCB} = \{g \in h/U: g^{-1} \sigma(g) \in U \sigma(U)\}$

$\hookrightarrow$
 $h(K) \xrightarrow{\pi} S(K) = K \times S_M(K)$

$\overline{Y_{SCB}} = Y_{SCB}/K$

$\hookrightarrow$
 $Y_{S_M \subset B_M} = \{m \in M/U_M: m^{-1} \sigma(m) \in U_M \sigma(U_M)\}$
 $\hookrightarrow$
 $M(K) \xrightarrow{S_M(K)}$

$$R_{SCB}^4 = R\Gamma(Y_{SCB}, \bar{\mathcal{O}}_e[...]) [4], \quad R_{S_M \subset B_M} = R\Gamma(Y_{S_M \subset B_M}, \bar{\mathcal{O}}_e[...])$$

Th As $h(k)$ -rep, $R_{SCB}^4 \simeq \text{int}(R_{S_M \subset B_M}) \otimes WH_4$.

More precisely, $\theta: S_M(k) \rightarrow \bar{\mathcal{O}}_e^*$.

$$R_{SCB}^4[\theta] = \text{int}(R_{S_M \subset B_M}[\theta]) \otimes WH_4, \quad \text{where } \zeta: S_M(k) \rightarrow \{\pm 1\} \text{ is as in Tasho's talk.}$$

Sgn char $\zeta: S \curvearrowright L \subset V$, $\zeta \in X^*(S)$ weight of S_K

$$\{(\alpha, d+1)\}$$

- α asym. if $\Gamma \cdot \alpha \neq \Sigma \cdot \alpha$

$$\downarrow$$

$$\Gamma = \langle \sigma \rangle \subset \Sigma = \Gamma \times \{\pm 1\}$$

- α sym. if $\Gamma \cdot \alpha = \Sigma \cdot \alpha$

$$K \supset K_{\{\pm \alpha\}} \supset K$$

$$\zeta(s) = \prod_{\alpha \in \Sigma_{\text{sym}}/\Sigma} \text{sgn}^\sharp(\alpha(s)) \prod_{\alpha \in \Sigma_{\text{asym}}/\Sigma} \text{sgn}(\alpha(s))$$

Idea of the proof $G = HM$

$$\pi^{-1}(m) = \{hm \in G/U : m^{-1}h^{-1}\sigma(h)\sigma(m) \in U\sigma(u)\}$$

$$\frac{m^{-1}h^{-1}\sigma(h)m}{m} \cdot \frac{m^{-1}\sigma(m)}{m}$$

$$\stackrel{m^{-1}(-)m}{\simeq} \left\{ h \in H/m(L) : h^{-1}\sigma(h) \in m(L)\sigma(m(L)) \right\}$$

$$\uparrow \text{ HDL for } m(L)$$

$\dim m(L) \cap \sigma(m(L))$ does not depend on m .

$$R_{SCB}^4 = R\Gamma(Y_{S_M \subset B_M}, R\pi_* \bar{\mathcal{O}}_e [4])$$

$$d = \dim \left(\frac{m(L)}{m(L) \cap \sigma(m(L))} \right)$$

$$R^d \pi_* \bar{\mathcal{O}}_e [4]_m \stackrel{\text{non-can}}{\simeq} WH_4, \quad R^i \pi_* \bar{\mathcal{O}}_e [4]_m = 0, \quad i \neq d$$

Prop $R^d \pi_* \mathcal{O}_E[+]$ is a constant sheaf on $Y_{S_M \subset B_M}$.

Step 2. Understand the $G(k)$ -equiv. str.

X var. $\ni \Gamma$ finite group. Suppose Γ acts transitively on $\pi_0(X)$.

V vector space $\rightsquigarrow \underline{V}$ constant sheaf on X . Fix a component $C \subset X$.

Γ -equiv. str. on $\underline{V} \simeq \Gamma_C$ on V , $\Gamma_C = \text{Stab}_\Gamma(C)$

Suppose

- $Y_{S_M \subset B_M}$ is conn'd.

$G(k)$ -equiv. str. $\leftrightarrow$ rep of $G(k)$ on $R^d \pi_* \mathcal{O}_E[+]$.

- B_M rat'l, L rat'l.

$G(k)/B(k)$

$\downarrow$

$M(k)/B_M(k)$

classical situation.

$\xrightarrow{\quad} \overline{\quad}$

$T = S / \bar{k}$

$\frac{T}{T} \xleftarrow{\epsilon} \frac{B}{B} \xrightarrow{\gamma} \frac{G}{G}$

L char sheaf on T_M

$\downarrow \pi$

$\frac{T_M}{T_M} \xleftarrow{\epsilon_M} \frac{B_M}{B_M} \xrightarrow{\gamma_M} \frac{G_M}{G_M}$

character for Weil-Haisenberg
 $\downarrow$
rep on $\frac{G}{G}$

Th. $\Gamma! q^* (AS_\psi \boxtimes I) \simeq \pi^* ((\Gamma_M)! q_M^* (L \boxtimes)) \otimes WH_\psi$

Lecture 4 (CC) ^{case} $(S \hookrightarrow G, \phi, \chi \in A(S) \cap B(G, F))$

famly
rem./F

$S(F) \rightarrow \mathbb{A}^x$

G -generic of depth r

Prop 4

Task. Construct an irrep of $G_{X,0}$. Further simplifies. just get sth. on $S_0 G_{X, \frac{r}{2}}$.

Strategy Begin by extending ϕ as much as possible to $S_0 G_{X, \frac{r}{2}}$

- ϕ extends to a char $\hat{\phi}: S_0 G_{X, \frac{r}{2}+} \rightarrow \mathbb{C}^\times$

- If $S_0 G_{X, \frac{r}{2}+} = S_0 G_{X, \frac{r}{2}}$, then we are done.

- If $S_0 G_{X, \frac{r}{2}+} \neq S_0 G_{X, \frac{r}{2}}$, can't try to induce $\text{Ind}_{S_0 G_{X, \frac{r}{2}+}}^{S_0 G_{X, \frac{r}{2}}}(\hat{\phi})$ is not irred.

Fact is a sum of $\sqrt{|V|}$ irreps,
each of dim $\sqrt{|V|}$.

$$\frac{S_0 G_{X, \frac{r}{2}}}{S_0 G_{X, \frac{r}{2}+}} \simeq \bigoplus_{\alpha \in \Phi_{\frac{r}{2}} \subset \Phi} \mathfrak{g}_\alpha$$

$\therefore V$ even dim'd

a) S split, can choose $\hat{B} \subset \mathcal{G}_{\mathbb{F}}$. This gives a way to split V into halves

$\text{Ind}_{S_0 G_{X, \frac{r}{2}+} \cup \frac{r}{2}}^{S_0 G_{X, \frac{r}{2}}}(\hat{\phi})$ is irred.

b) More generally, S unram. Can find $B \supset S_{\text{Fur}}$.

$$X^\pm = \left\{ g \in S_0 G_{X, \frac{r}{2}, \mathbb{F}} : g^{-1} \sigma(g) \in S G_{X, \frac{r}{2}, \mathbb{F}} \cup X, \frac{r}{2} \right\}$$

$$R^\pm(\phi) = \sum (-1)^i H_c^i(X^\pm, \mathbb{Z}\phi)$$

Rem This isn't the defin of pos. depth DL var. def by Laszlo's 2004.

Tuesday

$$L_{\mathbb{Q}}^{-1} \left(\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right)$$

Friday

$$L_{\mathbb{Q}}^{-1} \left(\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right)$$

$\uparrow$ $\mathbb{I}_h$ (Chen-Stasinski,
Ivanov-Nie)

ex $G = GL_2$,

$$\begin{pmatrix} \mathcal{O}^X & \mathcal{O} \\ \omega^{r+1} & \mathcal{O}^X \end{pmatrix}$$

Tues = Fri

$$\begin{pmatrix} \mathcal{O}^X & \omega^{[\frac{r+1}{2}]} \\ \omega^{[\frac{r+1}{2}]} & \mathcal{O}^X \end{pmatrix}$$

$$(1, \omega) \text{-conj.} \rightarrow \begin{pmatrix} \mathcal{O}^X & \omega \\ \omega^r & \mathcal{O}^X \end{pmatrix} \rightarrow \dots \rightarrow$$

c) Say S splits over E/F , tamely ram, not unram.

Key Lemma If $\alpha \notin \Phi_{r/2}$, then (inertia orbit of α) $\neq \alpha$.

$\implies \Phi_{\frac{r}{2}}$ can be broken up into paired inertia orbits.

$\rightsquigarrow$ yields a subspace $L \subset V$, stable under inertia & $S(\mathcal{O}_F)$

$\therefore$ def "over $\mathbb{F}_q$ "

General repeating c) for each step $\frac{G_{X,0}^i G_{X,\frac{r}{2}}^{i+1}}{G_{X,0}^i G_{X,\frac{r}{2}+}^{i+1}}$ produces an $S(F)$ -stable subgroup $K_{1,\frac{r}{2}}$

def over $\mathcal{O}_{F^+}$ which sits in the middle of K_1 & K_2 .

$$(T \hookrightarrow G, \phi, x) \rightsquigarrow (T \hookrightarrow G^0 \hookrightarrow G^1 \dots \hookrightarrow G^d, x) \quad , s_i = \frac{r_i}{2}$$

$$\phi_{-1} \quad \phi_0 \quad \phi_1 \quad \phi_d$$

$$0 = r_{-1} \leq r_0 < r_1 < \dots < r_d$$

$$K = \prod_{i=0}^d G_{X, s_{i-1}}^i$$

$$\nabla$$

$$K_1 = \cancel{G_{X,0}^0} \prod_{i=1}^d G_{X, s_{i-1}}^i$$

$$G_{X,0}^0 = \mathcal{O}^+$$

$$\supset K_{1,\frac{r}{2}}$$

$$\nabla$$

$$K_2 = \prod_{i=0}^d G_{X, s_{i-1}+}^i$$

~~L_i 's~~
a bunch of V_i 's

$$\nabla$$

$$K_3 = \ker(\hat{\phi})$$

Def (type DL) $X^\pm = \{ g \in K : g^{-1} \sigma(g) \in (B_{X,0}^0 \setminus K_{1,\frac{r}{2}}) / G_{X,r+} \}$

$$R^\pm(\phi) = \sum_{i \geq 0} (-1)^i H_c^i(X^\pm, \mathbb{Z}[\phi])$$

Th ($C(K, \mathbb{Z})$) $R^\pm(\phi) = \text{explicit recipe} = \text{Kir- Yu type assoc. to } (T \hookrightarrow G, \phi, \mathbb{Z})$

Art (cartoon)

$$H_c^* \left(L_{\tilde{G}}^{-1} \left(\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} \right), L\phi \right)$$

$$\phi_{-1} \otimes \phi_0 \otimes \phi_1 \otimes \phi_2$$

$$= H_c^* \left(L_{\tilde{G}}^{-1} \left(\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} \right), L\phi_{-1} \otimes \phi_0 \otimes \phi_1 \right) \otimes \phi_2$$

$$\cong H_c^* \left(L_{\tilde{G}}^{-1} \left(\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} \right), L\phi_{-1} \otimes \phi_0 \otimes \phi_1 \otimes \phi_2 \right) \otimes \phi_2 \otimes \text{WH}\psi_1$$

Key: push to smaller one is a constant sheaf w equiv. str. given by $(\xi \otimes \bar{\phi}_e) \otimes \text{WH}\psi$



Character sheaves on parahoric subgrps

$$G_{X,0;r+}$$

* T unram.

Conj (Lusztig 2006) If L is G -gen mult. loc. sys on

$T_{0;r+}$, then $\delta_! p^* L$ is simple perverse.

$$\frac{B_{0;r+}}{B_{0;r+}}$$



$$\frac{T_{0;r+}}{T_{0;r+}}$$

$$\frac{G_{0;r+}}{G_{0;r+}}$$

Th (Lusztig 2017, ²⁰²⁴ Bogan-Lusztig, C.) True
 $r=1,3$ arb. r

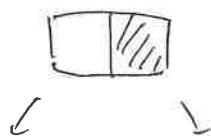
$$\text{Th. (2024 Bogan-Lusztig - C.) } \pi_{1,f}^* : D_{M_{0;r+}}^{\psi} \rightarrow D_{G_{0;r+}}^{\psi}$$

t-exact equiv.

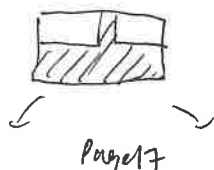
In particular, for any "geom. Yu data", can produce a simple pervers. sh. on $G_{0;r+}$

↳ @ depth 0, can put any char. sh. in the sense of Lusztig

²⁰¹⁷
Th (Dongkuan Kim, T ; Ivanov - Nie - Zhihang Yu, M)



=



Def (CKZ, "type sheaves") Have a defn using horocycle transform.

Th (CKZ) type sh. version of type DL th.

(eg. uses the th at the end of Xinwen's talk)

$$\begin{array}{ccc}
 G = H \rtimes M & \xrightarrow{B} & \\
 \swarrow \frac{T}{T} & & \searrow \frac{G}{G} \\
 & &
 \end{array}
 \quad \text{v.s.} \quad
 \begin{array}{ccc}
 & \xrightarrow{\frac{B_M}{B_M}} & \\
 \swarrow \frac{T_M}{T_M} & & \searrow \frac{M}{M} \oplus (\text{wh twist}) \\
 & &
 \end{array}$$